

CLIENT DELIVERABLE

Auto-Scheduling Optimizer — Implementation Plan

Systack Documentation

ROLE	COLOR	HEX
Headers, CTAs	Navy	 #001a2d
Navy Light	Navy Light	 #002845
Secondary accents	Teal	 #007da9
Primary buttons, links	Cyan	 #00a1db
Gradients	Cyan Bright	 #00c5e0
Backgrounds	Gray 50	 #f8fafc
Cards	Gray 100	 #f1f5f9
Borders	Gray 200	 #e2e8f0
Muted text	Gray 400	 #94a3b8
Body text	Gray 600	 #475569
Headings	Gray 800	 #1e293b
Success	Green	 #22c55e
Error	Red	 #ef4444
Accent highlights	Purple	 #8b5cf6

Auto-Scheduling Optimizer — Implementation Plan

Automation ID: scheduling-optimizer

Version: 1.0

Status: Draft — Planning

Date: 2026-06-16

Builder: SOL / Assembly

1. Build Phases

Phase 1: Calendar Integration (3 hours)

Tasks:

- 1. Connect to booking calendar (Google Calendar or local DB)
- 2. Fetch all slots for next 48 hours

3. Normalize to standard slot format (start_time, end_time, status)

Phase 2: Gap Detection Algorithm (2 hours)

Tasks:

```
// Gap detection logic
const slots = getSlots(next48Hours);
const gaps = [];

for (let i = 0; i < slots.length - 1; i++) {
  const gapMinutes = minutesBetween(slots[i].end, slots[i+1].start);

  if (gapMinutes >= 30) {
    gaps.push({
      start: slots[i].end,
      end: slots[i+1].start,
      duration: gapMinutes,
      discount: gapMinutes >= 60 ? 30 : 20, // % off
      type: 'between_bookings'
    });
  }
}

// Also detect: completely empty hours
const emptyHours = findEmptyBlocks(slots);
for (const block of emptyHours) {
  if (block.duration >= 60) {
    gaps.push({
      start: block.start,
      end: block.end,
      duration: block.duration,
      discount: 40,
      type: 'empty_block'
    });
  }
}
```

Phase 3: Discount Slot Creation (2 hours)

Tasks:

1. Create discount booking slots from detected gaps
 2. Apply discount percentage to standard pricing
 3. Mark slots as “discount” type in booking system
 4. Set auto-expiry: slots expire 2 hours before start if unfilled
-

Phase 4: Customer Notification (2 hours)

Tasks:

1. Query past customers (CRM Lite) who book at similar times
2. Send targeted notifications:
 - Email: “Flash availability — [service] at [time] for [discount]% off!”

- SMS (optional): “Last-minute opening today at 2 PM — 30% off. Book now: [link]”
3. Include one-click booking link
 4. Limit: don’t notify same customer more than 2x/week

Phase 5: Fill-Rate Tracking (1 hour)

Tasks:

1. Track: gaps detected, discount slots created, slots filled
2. Calculate fill rate and revenue recovered
3. Adjust discount percentages based on fill rate:
 - Fill rate < 30% → increase discount
 - Fill rate > 70% → decrease discount (maximize revenue)

2. Timeline

PHASE	EST. HOURS
Phase 1	3
Phase 2	2
Phase 3	2
Phase 4	2
Phase 5	1
Total	10 hours (~2 weeks)

3. Success Metrics

METRIC	TARGET
Schedule utilization	> 90%
Discount slot fill rate	> 40%
Revenue from filled gaps	> \$500/month
Customer notification response	> 5% click rate

Specification document — not yet built.

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